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Current knowledge of detoxification mechanisms of xenobiotic in honey bees

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Abstract The western honey bee *Apis mellifera* is the most important managed pollinator species in the world. Multiple factors have been implicated as potential causes or factors contributing to colony collapse disorder, including honey bee pathogens and nutritional deficiencies as well as exposure to pesticides. Honey bees' genome is characterized by a paucity of genes associated with detoxification, which makes them vulnerable to specific pesticides, especially to combinations of pesticides in real field environments. Many studies have investigated the mechanisms involved in detoxification of xenobiotics/pesticides in honey bees, from primal enzyme assays or toxicity bioassays to characterization of transcript gene expression and protein expression in response to xenobiotics/insecticides by using a global transcriptomic or proteomic approach, and even to functional characterizations. The global transcriptomic and proteomic approach allowed us to learn that detoxification mechanisms in honey bees involve multiple genes and pathways along with changes in energy metabolism and cellular stress response. P450 genes, is highly implicated in the direct detoxification of xenobiotics/insecticides in honey bees and their expression can be regulated by honey/pollen constitutes, resulting in the tolerance of honey bees to other xenobiotics or insecticides. P450s is also a key detoxification enzyme that mediate synergism interaction between acaricides/insecticides and fungicides through inhibition P450 activity by fungicides or competition for detoxification enzymes between acaricides. With the wide use of insecticides in agriculture, understanding the detoxification mechanism of insecticides in honey bees and how honeybees fight with the xenobiotis or insecticides to survive in the changing environment will finally benefit honeybees' management.

Keywords Detoxification genes · P450 genes · Function · Metabolism pathways · Nutrition · Honeybee health management

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Introduction

Honey bees as pollinators, along with bumble bees and solitary bees, contribute to 35 % of the world food crop production making them one of the most prominent and economically important group of pollinators worldwide (Velthuis and van Doorn 2006; Klein et al. 2007). Bees also play an important ecological role by providing pollination services to wild plants, including those in Europe where 80 % need insects for pollination (Kwak et al. 1998). However, the decline of the honey bee (*Apis mellifera*) has been reported in several parts of the world over the last decades generating great concern (Biesmeijer et al. 2006; National Research Council of the National Academies 2007; Oldroyd 2007; Stokstad 2007; Goulson et al. 2008; VanEngelsdorp and Meixner 2010). It is proposed that the population of honey bees in the environment is influenced by multiple factors, including pathogens, parasites, deficiency in nectar resources, pollutants (Decourtye et al. 2010; Neumann and Carreck 2010), and especially pesticides (Henry et al. 2012). It is likely that multiple interacting stressors from parasites, pesticides, and lack of flowers are driving honey bee colony losses and declines of wild pollinators (Goulson et al. 2015).

Honey bees are exposed to phytochemicals through the nectar, pollen and propolis consumed to sustain the colony (Di Pasquale et al. 2013). They may also encounter mycotoxins produced by *Aspergillus fungi* infesting pollen in beebread (Foley et al. 2014; Johnson 2015). Moreover, honey bees can be exposed to in-hive acaricides, including the pyrethroid tau-fluvalinate and the organophosphate coumaphos, which have been used widely for control of *Varroa*, a devastating parasitic mite of honey bees (National Research Council of the National Academies 2007). The fact that tau-fluvalinate and coumaphos residues in hives have some negative effects on honey bees has been reported in many studies (Teeters et al. 2012; Frost et al. 2013; Williamson and Wright 2013; Zhu et al. 2014a). In addition to the acaricides, honey bees may also be exposed to antimicrobial drugs applied by beekeepers to control bacterial and microsporidial pathogens (Johnson et al. 2013), and plant protection products such as fungicides applied to flowers and flowering crops (Mussen 2008; Mullin et al. 2010; Barmaz et al. 2012). More importantly, exposure to these kinds of chemicals is very common with negative consequences, for example, Insect Growth-Regulating (IGR) Insecticides , such as diflubenzuron and fenoxycarb have also been reported to adversely affect honeybees in many cases (Thompson et al. 2005; Milchreit et al. 2016). Meanwhile, since the introduction of imidacloprid in the early 1990s, the use of different neonicotinoid insecticides (NIs) has grown considerably. They are used extensively for the control of important agricultural crop pests by spraying and also are widely used in seed dressings and soil additions (Blacquière et al. 2012). Residues of these systemic insecticides can be found in the plant pollen and nectar (Blacquière et al. 2012) and Clothianidin and thiamethoxam were the most frequently detected NIs in honey and pollen found in central Saskatchewan, Canada (Codling et al. 2016). Potentially, bees could be exposed on a large scale to NIs residues originating from crop seed dressings (Blacquière et al. 2012). The toxic effect of neonicotinoid insecticides on honey bees have been widely studied (Stark et al. 1995; Nauen et al. 2001; Guez et al. 2003; Iwasa et al. 2004; Wehling et al. 2009; Han et al. 2010, 2012; Alkassab and Kirchner 2016). Neonicotinoid insecticides not only have negative effects on reproduction (Decourtye et al. 2005; Abbott et al. 2008; Gregorc and Ellis 2011), and behavior traits such as communication, homing, foraging and olfactory learning (Desneux et al. 2007; Aliouane et al. 2009; Tan et al. 2015; Henry et al. 2012; Williamson et al. 2014; Karahan et al. 2015), and the immunity system of bees (Di Prisco et al. 2013; Mason et al. 2014; Brandt et al. 2016), but also decrease the whole population level of honey bees (Gill et al. 2012) and may put a colony at risk of collapse (Henry et al. 2012). However, our knowledge of honey bee detoxification mechanisms on NIs is limited. It is known that pesticides interact with other stresses of honey bees including acaricides, fungicides, antimicrobial drugs and pathogens, which in turn augment the adverse effect on honey bees (Gregorc et al. 2012; Johnson et al. 2013; Zhu et al. 2014a; Aufauvre et al. 2014; Dussaubat et al. 2016). It is reported that pesticide exposure can alter both detoxification gene expression pathways and immune responses, interfering with the health of individual honey bees (Boncristianai et al. 2012), and rendering bees more susceptible to parasites (Pettis et al. 2012, 2013; Goulson et al. 2015). Vice versa, infections by pathogens, virus or mites modulate the detoxification pathways and immune system, making honey bees more sensitive to toxins (Vidau et al. 2011; Goulson et al. 2015). However, there are also documents which stated that diet based on phytochemicals in honey/pollen reduced the sensitivity of honey bee to some insecticides (Mao et al. 2013; Schmehl et al. 2014). Therefore, understanding the detoxification mechanisms of honey bees to xenobiotics/pesticides and how pesticides influence honey bee detoxification pathways provide benefits in the management of honey bee health.

As one of the insects, honey bees have detoxifying systems which function in the metabolism of endogenous compounds and xenobiotics such as drugs, pesticides, plant toxins, chemical carcinogens and mutagens as well as other pest insects (Feyereisen 2005, 2012). However, honey bees have a deficit of detoxification genes spanning Phase I (functionalization), II (conjugation) and III (excretion) gene families (Berenbaum and Johnson 2015). The major enzyme super families responsible for the metabolism or detoxification of toxins are the cytochrome P450 monooxygenases (P450s) (Phase I), glutathione transferases (GSTs) (Phase I or II) and carboxylesterases (COEs) (Phase I) (Feyereisen 2005). Moreover, multidrug resistance proteins and other ATP-binding cassette transporters, which function in transport of Phase II conjugates out of cells for excretion, are proteins involved in Phase III detoxification (Dermaauw and Van Leeuwen 2014). Generally, several studies have demonstrated the involvement of the P450-, GST- or COE-enzyme families in pesticide and secondary metabolite detoxification in honey bees using different approaches (Claudianos et al. 2006; Johnson et al. 2006; Mao et al. 2011). Studies investigating the mechanisms involved in detoxification of pesticides in honey bees, for instance, have initially utilized enzyme assays or toxicity bioassays in the presence or absence of known inducers or inhibitors of specific enzyme families involved in detoxification (Iwasa et al. 2004; Johnson et al. 2006). With the development of quantitative real-time PCR (qRT-PCR), microarray and RNA-seq analysis, studies based on systems biology approaches, such as genome-wide transcriptome analysis of specific tissues or transcriptomic analysis of selected detoxification, immune and stress response genes after pesticide or xenobiotic exposure, allowed specific genes responsible for xenobiotic detoxification or genes related to detoxification pathways in honey bees to be easily defined. However, mRNA expression profiles do not always correlate with protein concentrations and metabolic state (du Rand et al. 2015). Integrated proteomic and metabolomic approaches to attain a global overview of the response of honey bees to chemicals have also been reported (du Rand et al. 2015). Finally, functional study techniques such as in vitro protein expression and metabolism assays (enzyme assays) allowed the confirmability of the function of the specific genes responsible for metabolizing chemicals/insecticides or their regulation. This review focuses on the journey or methods toward characterization of the mechanisms in detoxification of xenobiotic in honey bees. Because honey bees are vulnerable to synergistic interactions among xenobiotics proven by some studies (Johnson et al. 2013) and phytochemical content in honey or pollen also has a pronounced influence on tolerance of toxic compounds (Mao et al. 2011; Schmehl et al. 2014), we also discussed the underlying detoxification mechanism of how P450s involved in the synergistic interactions among xenobiotics and clarified the effect of nutrition on detoxification and P450 gene regulation in honey bees, which can be implicated in honeybee health management.

Bioassay and biochemical approach-revealing the detoxifying enzymes involved in xenobiotic detoxification

Using bioassay and enzyme tests, many studies initially demonstrated the role of metabolic detoxification in the survival of honey bees exposed to insecticides or toxics, especially the great role of P450s (Iwasa et al. 2004; Johnson et al. 2006, 2012). For instance, Johnson et al. (2006) examined the effects of piperonyl butoxide (PBO), S,S,S tributylphosphorotrithioate (DEF), and diethyl maleate (DEM) on the toxicity of three pyrethroids, cyfluthrin, lambda-cyhalothrin and tau-fluvalinate. They found that the toxicity of the three pyrethroids to bees was greatly synergized by the P450 inhibitor PBO and synergized at low levels by the carboxylesterase inhibitor DEF. Little synergism was observed with DEM. These results suggested that metabolic detoxification, especially that mediated by P450s, contributes significantly to honey bee tolerance of pyrethroid insecticides. Most importantly, the most potent synergism between tau-fluvalinate and PBO suggests that P450s are especially important in the detoxification of this pyrethroid and also explains why honey bees tolerate tau-fluvalinate more compared to the other two pyrethroid insecticides. Also, a synergism study using P450 inhibitor suggested that P450s are an important mechanism for cyano-substituted neonicotinoids (such as acetamiprid and thiacloprid) detoxification and contribute to low toxicity of these NIs to honey bees compared to the nitro substitution (such as imidacloprid) (Iwasa et al. 2004). Carboxylesterase, suggested to be involved in pyrethroid metabolism in honey bees (Johnson et al. 2006), was shown to be less involved in NIs detoxification compared to P450s by using an inhibition synergism study (Iwasa et al. 2004). ABC transporter involvement in honey bee xenobiotic detoxification is also suggested primarily by synergism studies utilizing inhibitors. The fact that the toxicity of three neonicotinoids and two acaricides is synergized by verapamil, which is the inhibitor of P-glycoprotein, points to multidrug resistance transporters in detoxification in honey bees (Berenbaum and Johnson 2015).

Besides that synergistic bioassay studies are designed to determine the importance of detoxification proteins such as P450s in insecticide resistance or tolerance other biochemical approaches such as enzyme activity assays and metabolism studies are combined to further confirm the role of detoxification proteins in honey bees resistant or tolerant to insecticides. For example, the biochemical analyses of the in vivo derivatives of tau-fluvalinate identified a fragment of 4′-hydroxylated tau-fluvalinate through comparison of chromatograms for the treatment without the P450 inhibitor PBO and the control with PBO (Mao et al. 2011). This specific fragment served as a marker indicating that P450s in the midgut metabolize tau-fluvalinate. In *A. mellifera*, induction of carboxylesterase activity by organophosphate, neonicotinoid, pyrethroid, phenylpyrazole, and spinosyn pesticides suggested the participation of COEs in xenobiotic metabolism (Carvalho et al. 2013). GST activity in larvae, pupae and nurse bees but not in foragers was induced by the pyrethroid flumethrin (Nielsen et al. 2000) and elevated GST and P450 activities were observed in workers consuming nectar and pollen of *Ziziphus jujube*, indicating the role of these enzymes in xenobiotic detoxification (Ma et al. 2011).

Transcript expression characterization-individual gene expression in response to xenobiotics exposure

Although detoxifying enzymes such as P450s or carboxylesterase could be linked to metabolism of toxins in honey bees through bioassay and biochemical approach, individual detoxifying genes or related genes in metabolic pathways is impossible revealed by these tests. Therefore, transcript expression levels of selected detoxification genes in response to xenobiotics identified by qRT-PCR were widely studied, which is helpful to find out the candidate detoxifying genes in response to xenobiotics. *CYP6AS1*, *CYP6AS3*, *CYP6AS4* are up-regulated by honey, pollen and propolis (Johnson et al. 2012); *CYP9Q1-3* is induced by their acaricide (tau-fluvalinate) and insecticide (cypermethrin and bifenthrin) substrates (Mao et al. 2011); *CYP6AS14* is up-regulated by the monoterpene thymol, used in-hive as an acaricide against varroa mites (Boncristianai et al. 2012). These up-regulated P450 genes in response to xenobiotics indicatetheir important role in detoxification in honey bees. Actually, up-regulation of CYP 6 and 9 subfamilies in resistant strains or in response to insecticides exposure has been widely associated with insecticide resistance (Müller et al. 2008; Gong et al. 2013) or insecticide tolerance in pest insects (Poupardin et al. 2008; Riaz et al. 2009). In the eastern honey bee *Apis cerana cerana*, the Sigma-class AccGSTS1 was up-regulated by phoxim, cyhalothrin and acaricide (Yan et al. 2013) and the Omega-class GST gene GSTO2 was induced by cyhalothrin, phoxim, pyridaben and paraquat (Zhang et al. 2013) and they might all be implicated in the stress response to insecticides (Claudianos et al. 2006). Sigma GSTs show low levels of activity with the typical GST substrates but they have a high affinity for the lipid peroxidation product 4-hydroxynonenal; Omega class GSTs may play key roles in metabolic processes (Claudianos et al. 2006).

Generally, the detoxification and stress responses of xenobiotics include three major and interrelated pathways: oxidation-reduction mediated by alcohol dehydrogenases, aldehyde dehydrogenases, cytochrome P450s, hydroxylases and peroxidases, etc., conjugation mediated by Glutathione S-transferases enzyme (GST) superfamily, etc., and hydrolysis catalyzed by carboxylesterases, etc (Xu et al. 2013). In addition to these three pathways, other pathways may be involved, such as those that involve ATP-binding cassette transporters (ABC transporters), cadherins, somerases, Lyases or heat shock proteins (Xu et al. 2013; Dermaauw and Van Leeuwen 2014; Berenbaum and Johnson 2015). While qRT-PCR analysis allows to get the expression profile of limited selecting detoxification genes, it does not increase our knowledge of the whole detoxification mechanism and pathways associated. Therefore, microarray and RNA-seq, transcriptome approaches need to be explored to determine genomic recourses on detoxification and target genes putatively involved in insecticide metabolism and tolerance (Bass et al. 2012). This approach can be useful for investigating the honey bee responses to toxic stressors by measuring all of the transcriptional changes induced by the insecticides, and explore all of the detoxifying genes and involved pathways with a global view.

Transcriptome analysis has been well performed in many pest insects in terms of insecticide resistance (David et al. 2005; Poupardin et al. 2010; Zhu et al. 2014b; Xu et al. 2015; Bajda et al. 2015). Xu et al. 2013 constructed a transcriptome of the detoxification and stress response genes expressed in a western North American bumble bee, *Bombus huntii*. They found *B. huntii* expressed at least 584 genes associated with detoxification and stress responses, including oxidation and reduction enzymes, conjugation enzymes, hydrolytic enzymes, ABC transporters, cadherins, and heat shock proteins. Moreover, they found that the mechanisms in different life stages and castes are different, with more genes and detoxification pathways identified in the adult females than in larvae, pupae, or adult males. However, they didn't examine the transcriptome analysis of *B. huntii* after exposure to a specific insecticide or chemical. The response of detoxification genes and pathways in bees would be different with exposure to different xenobiotics, in view of differential detoxification mechanisms involved in different insecticide resistance in pest insects. The genome of *A. mellifera* has been explored and only about half the numbers of xenobiotic detoxifying genes, including three superfamilies of detoxifying enzymes, exist compared to other insects such as *Drosophila melanogaster* and *Anopheles gambiae* (Claudianos et al. 2006). *A. mellifera* genome has similar numbers of detoxifying genes with *B. huntii* (Xu et al. 2013). The shortfalls of these detoxifying genes may contribute to the sensitivity of the honey bee to insecticides (Claudianos et al. 2006). However, this reduction in detoxification genes does not prevent bees from carrying out the enzymatic functions associated with these gene families (Yu et al. 1984; Johnson 2015). Examining the molecular bases of the detoxification mechanism of honey bees to insecticides or xenobiotics is very useful for the honey bee health management. Recently, transcriptome analysis has been used to examine honey bee response to insecticides or toxics. Schmehl et al. (2014) explored the impact of coumaphos and fluvalinate, the two most abundant and frequently detected pesticides in the hive, on genome-wide gene expression patterns of honey bee workers. They found significant changes in 1118 transcripts, including genes involved in detoxification, behavioral maturation, immunity, and nutrition. Their results demonstrated that pesticide exposure can substantially impact gene expression involved in several core physiological pathways in honey bee workers. Moreover, although the two pesticides have different modes of action, there was significantly more overlap than expected by chance in the molecular responses they elicited, with 117 genes commonly regulated by both coumaphos and fluvalinate. Overlaps include several detoxification genes including three P450s (*CYP9S1*, *CYP6AS3*, and *CYP6AS4*) from the CYP3 clade: this clade is commonly involved in insecticide detoxification (Claudianos et al. 2006), indicating that these three genes might play important roles in detoxifying coumaphos and fluvalinate in honey bees. Derecka et al. 2013 analyzed molecular profiles obtained from worker honey bees larvae of which hives were exposed to syrup tainted with a low dose of imidacloprid. Their results on genome-wide RNA transcriptional responses and lipid profiles provided insight into the physiological responses of this worker honey bees larvae confronted with a real field environmental threat. Within a catalogue of 300 differentially expressed transcripts in larvae from imidacloprid treated hives, they detected significant enrichment of genes functioning in lipid-carbohydrate-mitochondrial metabolic networks. RNA levels for a cluster of genes encoding detoxifying P450 enzymes were elevated, with coordinated down-regulation of genes in glycolytic and sugar-metabolizing pathways, indicating that the stress response to the insecticide, besides the direct detoxifying genes, included other genes with the whole physiology pathways changing significantly and concurrently in order for honey bees to cope with the stress for survivorship. Specifically, within the overexpressed group of 105 transcripts, they found enrichment for nine genes of the cytochrome P450, which belong to the CYP4, CYP6 and CYP9 clades. As we know, overexpression of genes coding for the P450 clades (CYP4, CYP6 and CYP9) contributes considerably to insecticide-resistance (Claudianos et al. 2006). However, expression of nine of these transcripts were unaffected by exposure to imidacloprid or fipronil in a subsequent study (Aufauvre et al. 2014). The difference of the response of P450 genes might be due to the different time and duration of the honey bees' exposure to the imidacloprid. In addition, it has been declared that some P450 genes in honeybees which are involved to detoxification insecticides are constitutively expressed instead of induction expression (Mao et al. 2011). CYP9Q1 and CYP9Q2 can metabolite tau-fluvalinate identified through a functional study, while the transcript levels of them are not affected by tau-fluvalinate (<0.5-fold induction) (Mao et al. 2011). Alptekin et al. 2016 conducted a function study to identify the detoxification capacity of some thiacloprid -inductive P450 genes and they found that these P450 genes induced by thiacloprid do not have the metabolism function for thiacloprid. Their result indicated that some constitutively expressed P450 genes in honeybees may contributed the detoxification of thiacloprid instead of some inductive P450 genes. Therefore, whether the P450 genes in 4, 6 and 9 sub-families identified in transcriptome analysis really are responsible for the detoxification of imidacloprid or other insecticides in honey bees, further functional analyses need to be conducted.

Proteomic approach-individual protein expression in response to xenobiotics exposure

The levels of protein expression encoded by genes have not been measured in some transcript level studies as mentioned above. We currently do not know how changes in RNA expression levels translate into protein levels (Derecka et al. 2013). It has been stated that the mRNA expression profiles do not always correlate with protein concentrations and metabolic state (du Rand et al. 2015) and interpretations of the mRNA expression results are therefore limited. Roat et al. 2015 conducted a toxico-proteome profiling of the brain of newly emerged and aged honeybee workers that were exposed to sublethal dose (10 pg per day during 5 days) of fipronil. Their results suggested that the expression pattern of certain proteins that are involved in pathogen susceptibility, neuronal chemical stress, neuronal protein misfolding, a higher occurrence of apoptosis, etc., were changed. However, their study didn't play more attention on the metabolic pathways of fipronil in honeybee workers after exposure to sublethal dose fipronil. To identify protein networks and metabolic pathways involved in the detoxification of nicotine, du Rand et al. (2015) employed an integrated proteomic and metabolomic approach to attain a global overview of the response of honey bees to a 3-day nicotine exposure using mass spectrometry-based comparative proteomic and metabolomic analysis. Finally, they proposed a detoxification mechanism in honey bees involving multiple proteins/genes and pathways, revealing complex metabolism networks. They found that consumed nicotine is oxidized to less toxic metabolites, cotinine and cotinine N-oxide. CYP6 or CYP9 enzymes and GSTD1 (glutathione S-transferase delta isoform 1) may contribute to the detoxification. GSTD1 is widely distributed in *A. mellifera* tissues (Corona et al. 2005) and Delta GSTs have been implicated in insecticide resistance in several insects including *B. tabaci* (Guo et al. 2014). To support or drive the detoxification processes, the increased energy production in terms of up-regulation of enzymes involved in ATP synthesis, sugar catabolism, glycolysis and the TCA (tricarboxylic acid) cycle were found. ROS (reactive oxygen species) production increased in tandem with the increased energy production, inducing enhanced expression of enzymatic and non-enzymatic antioxidants, specifically GSTS1 (glutathione S-transferase sigma isoform 1), Gtpx-2 (phospholipid-hydroperoxide glutathione Peroxidase), Txp-1 (peroxiredoxin) and vitellogenin to protect against oxidative damage. Other elements of the cellular stress response, specifically Hsp 90 (Heat shock protein, which increased 12-fold), are also substantially up-regulated to promote stress tolerance. The up-regulation of GSTs and Gtpx-2 lead to the concurrent up-regulation of anabolism of glutathione, the tripeptide substrate essential for the functioning of these enzymes. Finally, the increased synthesis of glutathione is supported by the precursors provided by the increased flux through the TCA cycle (du Rand et al. 2015). Therefore, their study revealed a complex response involving detoxification, oxidative and general stress response pathways concurrent with an increase in the insect's energy metabolism. Actually, that bees are capable of detoxifying nicotine in nectar has been reported before: bees fed sugar water containing 50 ppm nicotine produced honey with only 3.2 ppm nicotine (Singaravelan et al. 2006). It has been stated that imidacloprid in honey bees can be metabolized only by Phase I enzymes into five metabolites including 4/5-hydroxy-imidacloprid, 4,5-dihydroxy-imidacloprid, 6-chloronicotinic acid, imidacloprid olefin and urea derivatives (Suchail et al. 2004). Most C^{14}-labeled acetamiprid was metabolized within 30 min of dosing (Brunet et al. 2005) likely due to rapid cytochrome P450 detoxification as demonstrated by a synergy study (Iwasa et al. 2004). However, the detoxification genes involved in metabolic pathways have not been exposited. Proteomic and metabolomic analysis on detoxification mechanisms of honey bees to insecticides, especially NIs, has not been reported yet. The neonicotinoids (NIs) are synthetic analogs of nicotine with a mode of action resembling that of nicotine. Therefore, NIs may share similar detoxification pathways in honey bees as nicotine and need further investigation. Understanding how healthy honey bees process NIs under normal conditions will improve our understanding of how NIs, on their own or in synergy with other stress factors, lead to a decline in bee vitality (du Rand et al. 2015).

Function approach-pinpointing the function of the special genes involved in detoxification

With the availability of new functional study techniques such as double-stranded RNA-mediated gene interference (RNAi), in vitro protein expression using baculovirus expression system and transgenic gene expression, significant progress has been made in identifying the functions of the genes that are involved in insecticide detoxification or insecticide resistance at the whole genome level (Liu et al. 2015). Through transcriptome and proteomic analysis, the potential detoxification genes are discovered; however, it remains to be shown if altered transcription of these CYP/P450 genes is a specific detoxification-response and whether the encoded enzymes are capable of metabolizing insecticides. Mao et al. (2011) expressed CYP3 clan midgut P450s using a baculovirus-insect expression system and demonstrated that CYP9Q1, CYP9Q2, and CYP9Q3 metabolize tau-fluvalinate to a product that suitable for further cleavage by the carboxylesterases that also contribute to tau-fluvalinate tolerance. These in vitro assays indicated that all of the three CYP9Q enzymes also detoxify coumaphos. In their earlier study using the same method, Mao et al. (2009) demonstrated that CYP6AS1, CYP6AS3, CYP6AS4 and CYP6AS10 are capable of contributing to the detoxification of quercetin, a flavonol widely distributed in plant nectars.

To date, studies examining the response of honey bees to neonicotinoid exposure are limited, especially using the functional approach. The functional evidence of P450 to metabolize neonicotinoid insecticides has been reported in some other insects (Jouben et al. 2008; Karunker et al. 2009; Roditakis et al. 2011). For instance, *Drosophila melanogaster* Cyp6g1 was found up-regulated in a DDT-resistant strain (Daborn et al. 2001) and analysis of transgenic flies has shown that over-expression of this gene is capable of conferring imidacloprid resistance (Daborn et al. 2002). More directly, heterologous expression of CYP6G1 has shown that CYP6G1 can metabolize imidacloprid to its 4- and 5-hydroxy forms, thus implicated in exhibiting 8-fold cross-resistance to imidacloprid in *Drosophila* (Jouben et al. 2008). Several detoxifying genes such as P450 and GST have been implicated in the detoxification of imidacloprid in honey bees through transcript gene expression analysis (Derecka et al. 2013). Further experimental studies will be required to establish whether honey bees have a functional orthologue of the *D. melanogaster* CYP6G1, the CYP/P450 enzyme capable of protecting insects against the neonicotinoid insecticide.

Underlying detoxification mechanisms for synergy of xenobiotics in honey bees

Honey bees are forage insects and easily exposed to mixtures of xenobiotics applied to plants including fungicides, herbicides and insecticides in addition to very high levels of acaricides which may be present within colonies. Studies have demonstrated that the synergy effect of exposure to different xenobiotic combinations in honey bees has profound effects on honey bee colonies and may significantly contribute to honey bee colony loss (Schmuck et al. 2003; Iwasa et al. 2004; Johnson et al. 2009, 2013; Glavan and Bozic 2013). There are underlying reasons for the synergy effect between different xenobiotics in honey bee *A. mellifera*, well summarized by Glavan and Bozic 2013. Besides the mechanism on target site interaction between xenobiotics, the majority of synergistic effects observed in honey bees are ascribed to the inhibition of detoxifying midgut enzymes P450s involved in xenobiotic metabolism (Glavan and Bozic 2013). In spite of piperonyl butoxide (PBO), the classic inhibitor of P450s, which has been reported to be involved in the synergized toxicity of insecticides (tau-fluvalinate, coumaphos and neonicotinoid)

(Johnson et al. 2009, 2013), ergosterol biosynthesis inhibitor fungicides such as prochloraz, propiconazole, epoxiconazole, carbendazimand insecticides, due to their fungicide inhibitory action on P450s, have been demonstrated to elevate the toxicity of many insecticides including neonicotinoid insecticides (acetamiprid, thiacloprid, imidacloprid) (Schmuck et al. 2003; Iwasa et al. 2004), pyrethroid insecticides (deltamethrin, lambda-cyhalothrin and alpha-cypermethrin) (Thompson and Wilkins 2003; Glavan and Bozic 2013), and hive varroacides (coumaphos, flumethrin, fenpyroximate) (Thompson and Wilkins 2003; Johnson et al. 2013). The sterol biosynthesis inhibiting (SBI) fungicide prochloraz elevated the toxicity of the acaricides tau-fluvalinate, coumaphos and fenpyroximate, likely through inhibition of detoxicative cytochrome P450 monooxygenase activity (Johnson et al. 2013). Synergism between in-hive acaricides in honey bees were also documented (Johnson et al. 2009, 2013). For example, due to previous treatment with coumaphos, the toxicity of tau-fluvalinate to bees was largely increased (Johnson et al. 2009). The molecular basis of the synergism between in-hive miticides is the metabolism competition mediated by P450 enzymes at P450 catalytic sites (Glavan and Bozic 2013). Mao et al. (2011) found that coumaphos and fluvalinate dock into the same catalytic pocket of CYP9Q1, CYP9Q2 and CYP9Q3 demonstrated by molecular models. Sublethal amitraz pre-treatment increased the toxicity of the three P450-detoxified acaricides (tau-fluvalinate, coumaphos and fenpyroximate), but amitraz toxicity was not changed by sublethal treatment with the same three acaricides. This phenomenon may be because honey bees tolerate to amitraz not by the P450s (Johnson et al. 2013). Esterase was less used for explanation of the underlying detoxification mechanism for synergism between xenobiotics in honey bees, likely because esterase is less involved in detoxification in honey bees compared to P450s. Understanding the mechanisms of the effect of combination pesticides on honey bees is helpful so that beekeepers correctly use pesticide and bee-hive compounds to avoid the negative impact on honey bees. Therefore, there is a vital need for studies on potential interaction of pesticide combinations and the underlying detoxification mechanism for evaluating whether novel or existing compounds can be considered safe for honey bees.

The effect of nutrition on detoxification and P450 gene regulation in honey bees

As we know, honey bee colonies are highly dependent upon the availability of floral resources from which they get the nutrients (notably pollen) necessary to their development and survival (Di Pasquale et al. 2013; and see Huang 2012 for a review). It has been suggested that pollen ingested can not only influence the physiological metabolism (Alaux et al. 2011; Ament et al. 2011), immunity (Alaux et al. 2010), tolerance to pathogens like bacteria (Rinderer et al. 1974), virus (Degrandi-Hoffman et al. 2010) and microsporidia (Rinderer and Elliott 1977), but also can influence the sensitivity of honey bees to pesticides (Wahl and Ulm 1983). This effect may result in part because constituents in pollen or honey can regulate detoxifying genes expression and as a result affect the metabolic pathways of phytochemical and pesticides. It has been proposed that regulation of detoxification genes in *A. mellifera* differs in some respects from P450 regulation in other insects, in which phenobarbital is an important inducer of insect P450 transcription, while in *A. mellifera*, phenobarbital hardly induces P450 transcription expression (Mao et al. 2011). Johnson et al. 2012 found that feeding bees with phenobarbital had no effect on the toxicity of tau-fluvalinate; Similarly, no P450 induction, as measured by tau-fluvalinate tolerance, occurred in bees fed xanthotoxin, salicylic acid, or indole-3-carbinol, all of which induce P450s in other insects; however, only quercetin, a common pollen and honey constituent, reduced tau-fluvalinate toxicity. There are also other evidences that diet can modulate expression of detoxification genes as well as pesticides in *A. mellifera*. Schmehl et al. 2014 applied whole genome microarrays and explored the association between nutrition- and pesticide-regulated gene expression patterns and demonstrated that bees fed a pollen-based diet exhibit reduced sensitivity to a third pesticide, chlorpyrifos. They demonstrated that expression levels of several of the putative pesticide detoxification genes (CYP9S1 and CYP9Q3) induced by pollen-based diet identified in this study contribute to the reduced sensitivity of honey bees to the chlorpyrifos. Up-regulation of detoxification genes in response to pollen feeding were also found in other studies (Mao et al. 2011, 2013; Johnson et al. 2012). Transcripts of CYP9Q3, the gene which could metabolize tau-fluvalinate and coumaphos (Mao et al. 2011), are strongly induced by honey extract fed on by honey bees (Mao et al. 2011). Mao et al. (2013) used RNA-Seq analysis to investigate the gene expression profiles in the midguts of honey bees fed with p-coumaric acid-containing sugar candy, and conclude that all classes of detoxification genes as well as select antimicrobial peptide genes were up-regulated, including seven genes encoding phase I enzymes, four genes encoding phase II enzymes and two genes encoding phase III enzymes, and gene encoding abaecin (an antimicrobial peptide mediating immunity against bacteria). These authors provide more evidences that functional significance of P450 up-regulation induced by p-coumaric acid by showing that rates of substrate disappearance of coumaphos in midguts of honey bees was significantly increased when fed with p-coumaric acid. They also demonstrated that exposure to pinobanksin 5-methyl ether, and pinocembrin, which are phenolics present in honey, increased the expression of detoxification genes in honey bees (Mao et al. 2013). These results indicated that honey extracts or constituents could increase the ability of honey bees to metabolize the insecticides through regulation of detoxifying genes expression. Moreover, bees may also benefit from the presence of quercetin and other flavonoids in honey/pollen because of their antioxidant or antimicrobial activity (Johnson 2015). Schmehl et al. (2014) also proposed that some pesticides and components in pollen or honey may modulate similar molecular response pathways. In other words, there is substantial overlap in responses to pesticides and pollen/honey containing diets at the transcriptional level of the detoxification system in honey bees, which may have evolved through honey bees' interaction with phytochemicals. In other herbivores, induction of detoxification enzymes, especially P450s after consumption of small amounts of plant material contribute to the tolerance and adaptation to the allelochemicals of the plant (see Glendinning 2002 for a review). Pollen may have a similar priming response in the honey bee by triggering up-regulation of P450s and thereby improved resistance to pesticide exposure (Schmehl et al. 2014). Thus, providing honey bees and other pollinators with high quality nutrition may improve resistance to pesticides (Mao et al. 2013) and have important implications on honey bee management (Schmehl et al. 2014).

However, our understanding of the molecular mechanisms associated with P450 gene regulation including factor and receptors is extremely inadequate in honeybees. Alptekin et al. (2016) found a number of GPCRs are co-overexpressed with some P450 genes in honey bees after exposure to thiacloprid treatment, indicating the possibility that these GPCRs genes may play a role in triggering/regulating the enhanced transcription of the CYP/detox genes. In fact, recent work about GPCRs has suggested they may be involved in regulating over-expressed P450s in resistant mosquitoes, *Culex quinquefasciatus*, and housefly, *Musca domestica* (Li et al. 2013, 2014). Li et al. (2014) reported that knockdown of four GPCR genes by RNA interference both decreased resistance to permethrin and repressed the expression of four insecticide resistance related P450 genes in *C. quinquefasciatus*. The downstream factors like PKA genes in the GPCR regulatory pathway have also been explored in a next study in order to characterize the role of PKA genes in regulation of insecticide resistance-related P450 genes in *C. quinquefasciatus* (Li et al. 2015). It would be interesting to examine the role of some receptors like GPCRs and GPCR regulatory pathway in honey bee P450 gene expression responses to xenobiotics.

Conclusions

Honey bees have fewer numbers of detoxifying genes spanning three phase enzymes than other insects, making them more sensitive to most fungicides and some specific insecticides used in agriculture. Although most acaricides are not so toxic to honey bees, in combination with other insecticides or fungicides bees encounter, the toxicity increases dramatically in part because of the competition of detoxification or inhibition of P450 detoxification mechanisms. Knowing the changes of gene expression and protein expression after exposure to chemical compounds allows for a better understanding of the molecular basis of how pesticides affect the physiology of honey bees, which can lead to declines of honey bees, and how honey bees detoxify and deal with the stress response of chemical compounds enabling them to adapt to and survive in these changing environments. In conclusion, most of the effects of xenobiotics on honey bees revealed a complex response involving detoxification, oxidative and general stress response pathways concurrent with an increase in the insect's energy metabolism. Therefore, the detoxification mechanism is a complicated physiological process and involves many factors and pathways which can be explored by transcriptome, proteomic and metabolomic analysis. With the wide use of insecticides in agriculture, understanding the detoxification mechanism of insecticides by honey bees and how insecticides impair and influence the honey bee detoxification mechanism is very important. In our review we found that P450s play a greater role in detoxification of xenobiotics in honey bees rather than GST and COE. The P450 genes whose potential was proven in detoxification pathways are dominating in the CYP 4, 6 and 9 families, which are also involved in insecticides resistance in other pest insects. In addition, GST, especially Delta GSTs GSTD1, which is widely distributed in *A. mellifera* tissues may also contribute to the detoxification. However, the direct evidence of metabolism of specific xenobiotics or insecticides (especially neonicotinoid insecticides) by individual P450 genes is still under reported, although some studies have explored the functions of some P450 genes in CYP 9Q family in metabolism of acaricides, and need further investigation. The fact that P450 genes and immunity genes can be regulated by nutrition such as honey substitutes has implications in honey bee management. Honey substitutes like p-coumaric acid have been suggested as an additive to honey substitutes to allow beekeepers to maintain colonies during food shortages without compromising the ability of their bees to defend themselves against the pesticides and pathogens that currently bedevil beekeeping in the United States (Mao et al. 2013). Yet, our understanding of the molecular mechanisms associated with gene regulation including factor and receptors is extremely inadequate and future research should focus on the upstream factors in the P450 gene regulatory pathway in order to characterize their involvement in xenobiotics detoxification in honey bees. This is expected to support the effort to identify new genes that can regulate P450s expression and help build a better understanding of the regulatory pathway of xenobiotics/insecticide detoxification in honey bees. Finally, additional functional analyses of candidate responsive genes identified in previous studies need further investigation that may reveal important insights into transgenic engineering for foraging honey bee health management.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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